For the questions 1,2,3 consider the circuit shown in figure 1 and answer:

1) The equivalent resistance Req between the terminals a-b of the voltage source is: ..2..Ω

2) The current Io in the 6 Ω resistor is: ..2.23..A

3) The voltage (V) across 2 Ω resistor is: ..4.0..V

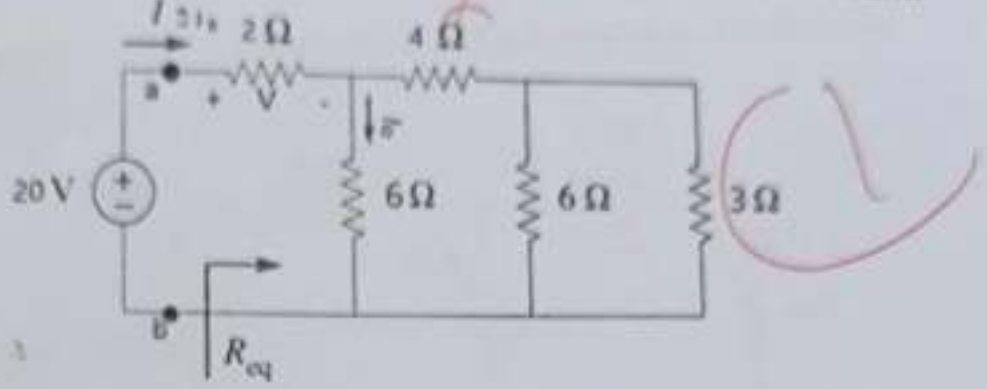


Fig.1

For the questions 4,5,6,7,8 consider the circuit shown in figure 2 and answer (use nodal analysis method):

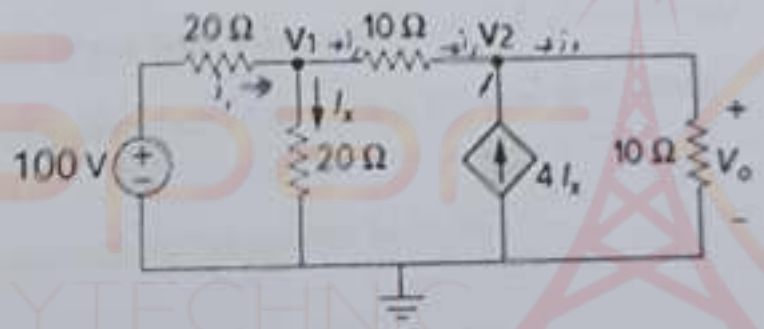


Fig.2

4) The value of node voltage V_1 is: ..45.45 V..

5) The value of node voltage V_2 is: ..27.27 V..

6) The value of the current (Ix) in 20 Ω resistor is: ..2.2725 A..

7) The power absorbed by 10 Ω resister is: ..74.36 W..

8) Write down the nodal equations (don't simplify):

$\frac{100 - V_1}{20} = \frac{V_1 - V_2}{10} + \frac{V_1}{20}$

$\frac{V_1 - V_2}{10} + 4I_x = \frac{V_2}{10}$

* For the questions 9,10,11, 12,13 consider the circuit shown in figure 3 and answer:

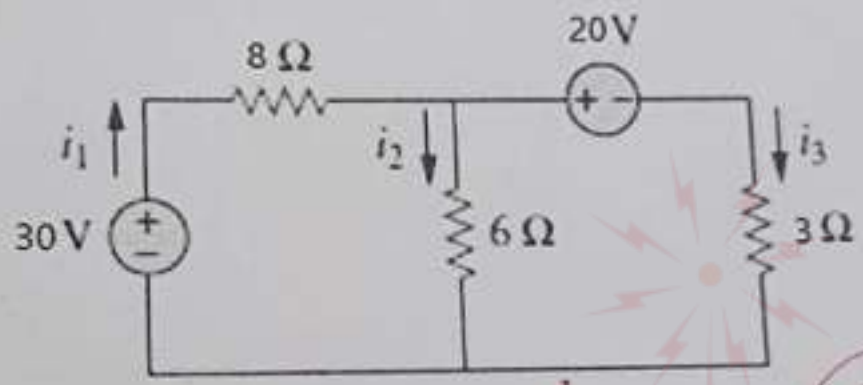


Fig.3

9) The current i1 is: $\frac{30}{8} = 3.75$ A

10) The current i3 is: 6.66 A

11) The delivered power by the 30 V voltage source is: 112.5 W

12) How many joules of energy will a 6 Ω resistor dissipate in 20 minutes ?

0.3.3.60 J

13) The total charge entering an 8 Ω resistor in 1h is:

1 $R_{eq} = 5\Omega$ (4)

2 $20 = 8I$ (2)

3 $V = 8V$ (2)

4. $V_1 = 100V$ (2)

5 $V_2 = 130V$ (2)

6. $I_x = 5A$ (2)

7. $(5^2) \times 10 = 250W$ (2)

8. 2250W -

(8)

$\frac{V_1 - 100}{20} + \frac{V_1 - V_2}{10} + \frac{V_1}{20} = 0$ (3) (12)

$\frac{V_2 - V_1}{10} + 4I_x + \frac{V_2}{10} = 0$ (2)

5 equations

(7)

9 - $I_1 = 1.666$ A (2)

10 - $I_2 = -1.115$ A (2)

11 → $P_{10} = 49.8W$ (1.4)

12 - $W = 55.944.1$ Joules (3)